

HDS06 - Designing and Querying a Clinical Trial Participant Registry

Westbrook University Hospitals NHS Trust, Clinical Trial Participant Registry

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Introduction

This report designs a relational database for the Westbrook University Hospitals NHS Trust Clinical Trial Participant Registry, implements it in Microsoft SQL Server (T-SQL), populates it, and answers seven clinical/research questions. The accompanying runnable script is `script/HDS_DB_LAINSBURY_2026.sql`.

Note

Reproducing the results. Execute the scripts in order (00 → 01 → 02 → 03 → 05 → 04 → 06) or run the single combined file `HDS_DB_LAINSBURY_2026.sql`; it runs without errors from start to finish. Each query returns one result grid, screenshotted below from the VS Code mssql extension connected to SQL Server. The *expected result* listed under each query states the output the script reproduces.

Chapter 1

Part 1: Database Design

1.1 Entity Relationship Diagram

The schema has eight tables. **TrialSite** resolves the many-to-many relationship between trials and sites; **Enrollment** records a participant's participation in a trial at a site; **ConsentSigning** is the append-only log of consent events (Figure 1.1).

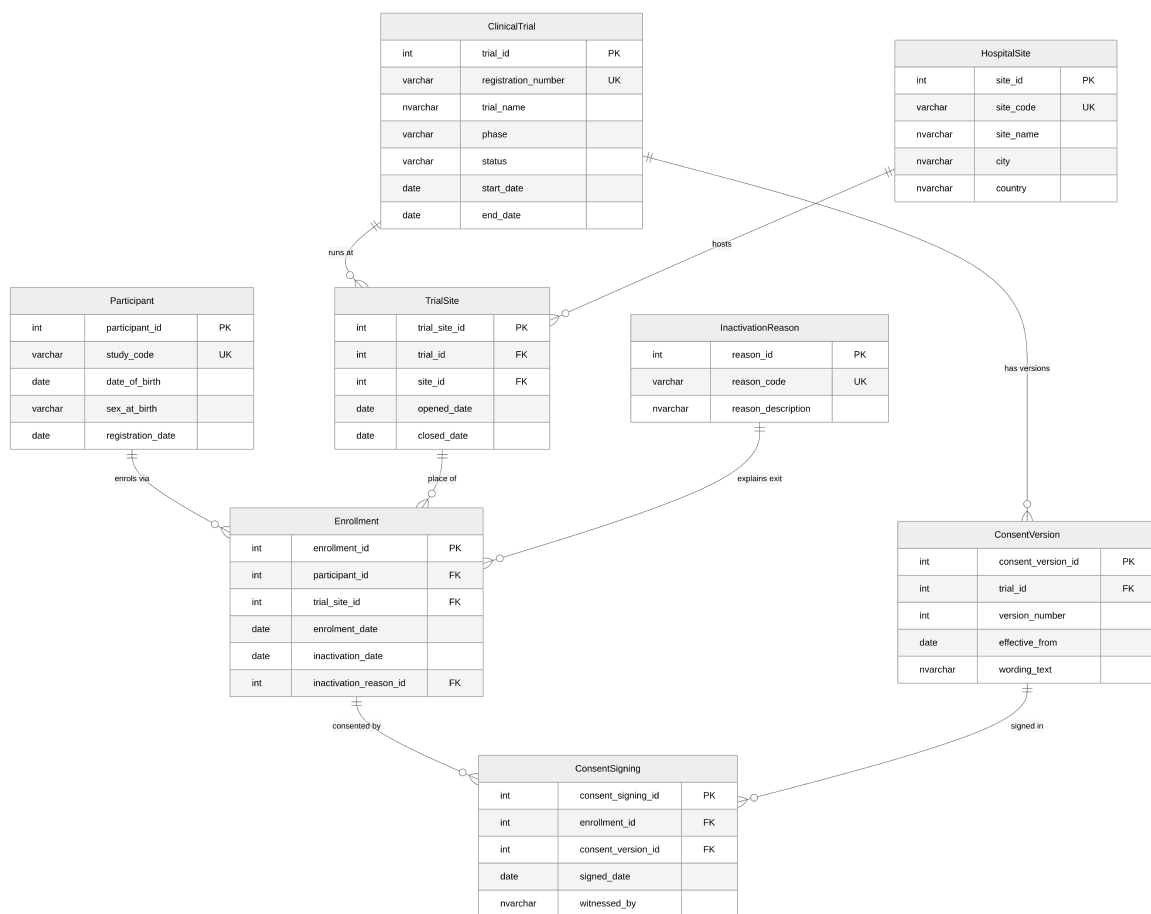


Figure 1.1: Entity Relationship Diagram (crow's-foot notation).

1.2 Table Descriptions

Every table uses a surrogate INT **IDENTITY** primary key for stable internal joins, plus one or more **business (natural) UNIQUE keys** for the values clinical staff actually recognise.

Table 1.1: Table descriptions and key design choices

Table	Purpose and key design choices
ClinicalTrial	One row per trial. PK trial_id (IDENTITY). UNIQUE registration_number — the ISRCTN number is the externally recognised, never-reused identifier, so it is the natural key. CHECK on phase (I/II/III/IV) and status enforce the controlled vocabularies; CHECK (end_date >= start_date) prevents impossible ranges.
HospitalSite	One row per site. PK site_id . UNIQUE site_code — short codes (e.g. WBK-GEN) are how sites are referred to operationally, so they form the natural key.
Participant	One pseudonymised participant per row; no names held . PK participant_id . UNIQUE study_code — the pseudonym is the only safe external identifier. CHECK on sex_at_birth ; CHECK (date_of_birth < registration_date).
InactivationReason	Controlled vocabulary of reasons a participant leaves a trial, so exits report consistently across trials. PK reason_id . UNIQUE reason_code — the stable report token.
TrialSite	Link table resolving the trial–site many-to-many; also records when each site opened/closed for recruitment on that trial. PK trial_site_id . UNIQUE (trial_id , site_id) is the real business key; FKs to ClinicalTrial and HospitalSite; CHECK (closed_date >= opened_date).
ConsentVersion	Every approved version of a trial’s consent form, including wording and effective date; old versions never overwritten. PK consent_version_id . UNIQUE (trial_id , version_number); FK to ClinicalTrial; wording_text is NVARCHAR(MAX).
Enrollment	A participant enrolled in a trial at a specific site. Points at a TrialSite row so trial and site are captured together and a participant can only enrol where the trial runs. Exit is recorded without deleting the row. PK enrollment_id ; FKs to Participant, TrialSite, InactivationReason; UNIQUE (participant_id , trial_site_id); paired-null CHECK; filtered unique index for “one active trial at a time”.
ConsentSigning	Append-only log of every consent signing event: when, which version, who witnessed. PK consent_signing_id ; FKs to Enrollment and ConsentVersion; UNIQUE (enrollment_id , consent_version_id); protected by an append-only trigger.

Check constraints. Phase/status/sex use CHECK ... IN (...) to guarantee clean categorical data for cross-trial reporting. Date checks (**end_date** >= **start_date**, **closed_date** >= **opened_date**, **inactivation_date** >= **enrolment_date**) prevent logically impossible records.

The paired-null check on **Enrollment** guarantees a leaving date never appears without a reason and vice-versa.

1.3 NULL and NOT NULL Decisions

Table 1.2: Nullability decisions and their justification

Column	Decision	Meaning of NULL / rationale
<code>ClinicalTrial.end_date</code>	NULL	NULL is meaningful: the trial is still running. Current trials have no end date <i>yet</i> .
<code>ClinicalTrial.start_date</code> , <code>status</code> , <code>phase</code>	NOT NULL	A trial cannot exist without these; absence would be an error, not a real state.
<code>TrialSite.closed_date</code>	NULL	NULL = the site is still open for recruitment on that trial.
<code>Enrollment.inactivation_date</code> & <code>inactivation_reason_id</code>	NULL (paired)	The classic “meaningful absence”: NULL = still active. Both fill in on exit; the paired-null CHECK keeps them in step.
<code>Participant.date_of_birth</code> , <code>registration_date</code> , <code>sex_at_birth</code>	NOT NULL	Recorded at registration and required for eligibility, age and demographic reporting.
<code>ConsentSigning.signed_date</code> , <code>witnessed_by</code>	NOT NULL	A consent event without a date or witness is not a valid consent event.
<code>ConsentVersion.wording_text</code> , <code>effective_from</code>	NOT NULL	A consent version is meaningless without its wording and the date it took effect.

1.4 Relationships Between Tables

Table 1.3: Relationships between tables

Tables	Type / FK	What it represents
ClinicalTrial – HospitalSite	M:N via TrialSite (FK <code>trial_id</code> , <code>site_id</code>)	A trial runs at many sites and a site hosts many trials; the link table also carries the open/close recruitment dates.
ClinicalTrial -> ConsentVersion	1:M (FK <code>trial_id</code>)	One trial accumulates many consent-form versions over its life as the protocol changes.
Participant -> Enrollment	1:M (FK <code>participant_id</code>)	A participant may have several enrolments over time, only one active at once (filtered index).
TrialSite -> Enrollment	1:M (FK <code>trial_site_id</code>)	Each enrolment happens at exactly one trial-site.

Tables	Type / FK	What it represents
InactivationReason -> Enrollment	1:M optional (FK <code>inactivation_reason_id</code>)	A standardised reason classifies many exits; active enrolments reference none.
Enrollment -> ConsentSigning	1:M (FK <code>enrollment_id</code>)	One enrolment can have several signing events (original + re-consent after amendment).
ConsentVersion -> ConsentSigning	1:M (FK <code>consent_version_id</code>)	Each signing records which version was signed; one version is signed by many participants.

Many-to-many resolution. The only many-to-many relationship (ClinicalTrial – HospitalSite) is resolved with the `TrialSite` junction table, which carries its own surrogate key, a **UNIQUE** (`trial_id`, `site_id`) business key, and the recruitment open/close dates that are attributes of the *pairing* rather than of either entity alone.

Chapter 2

Part 2: SQL

2.1 INSERT statements — one per table

All foreign-key values are resolved with subqueries on business keys — no hard-coded IDENTITY integers. The remaining INSERTs are in 02_insert_reference_data.sql and 03_insert_operational_data.sql.

ClinicalTrial

```
INSERT INTO dbo.ClinicalTrial (registration_number, trial_name, phase, status,
    ↪ start_date, end_date)
VALUES ('ISRCTN10234567', N'CARDIOPROTECT: Cardiac Rehabilitation in Post-MI
    ↪ Patients',
        'III', 'active', '2023-03-01', NULL);
```

HospitalSite

```
INSERT INTO dbo.HospitalSite (site_code, site_name, city, country)
VALUES ('WBK-GEN', N'Westbrook General Hospital', N'London', N'England');
```

Participant

```
INSERT INTO dbo.Participant (study_code, date_of_birth, sex_at_birth,
    ↪ registration_date)
VALUES ('WBK-001', '1958-04-12', 'Male', '2021-11-03');
```

InactivationReason

```
INSERT INTO dbo.InactivationReason (reason_code, reason_description)
VALUES ('consent_withdrawn', N'Participant withdrew their consent to take part
    ↪ in the trial');
```

TrialSite (FK look-ups via subquery)

```
INSERT INTO dbo.TrialSite (trial_id, site_id, opened_date, closed_date)
VALUES (
```



```
(SELECT trial_id FROM dbo.ClinicalTrial WHERE
↪ registration_number='ISRCTN10234567'),
(SELECT site_id FROM dbo.HospitalSite WHERE site_code='WBK-GEN'),
'2023-03-01', NULL);
```

ConsentVersion

```
INSERT INTO dbo.ConsentVersion (trial_id, version_number, effective_from,
↪ wording_text)
VALUES (
(SELECT trial_id FROM dbo.ClinicalTrial WHERE
↪ registration_number='ISRCTN10234567'),
1, '2023-03-01',
N'I agree to take part in the CARDIOPROTECT study. ... I may withdraw at any
↪ time without affecting my medical care.');
```

Enrollment (resolves the trial-site from registration number + site code)

```
INSERT INTO dbo.Enrollment (participant_id, trial_site_id, enrolment_date,
↪ inactivation_date, inactivation_reason_id)
VALUES (
(SELECT participant_id FROM dbo.Participant WHERE study_code='WBK-001'),
(SELECT ts.trial_site_id FROM dbo.TrialSite ts
JOIN dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
JOIN dbo.HospitalSite hs ON hs.site_id = ts.site_id
WHERE ct.registration_number='ISRCTN10234567' AND hs.site_code='WBK-GEN'),
'2023-03-15', NULL, NULL);
```

ConsentSigning (resolves enrolment + version by business keys)

```
INSERT INTO dbo.ConsentSigning (enrollment_id, consent_version_id, signed_date,
↪ witnessed_by)
VALUES (
(SELECT e.enrollment_id FROM dbo.Enrollment e
JOIN dbo.Participant p ON p.participant_id = e.participant_id
JOIN dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
JOIN dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
WHERE p.study_code='WBK-001' AND ct.registration_number='ISRCTN10234567'),
(SELECT cv.consent_version_id FROM dbo.ConsentVersion cv
JOIN dbo.ClinicalTrial ct ON ct.trial_id = cv.trial_id
WHERE ct.registration_number='ISRCTN10234567' AND cv.version_number=1),
'2023-03-15', N'Nurse A. Okafor');
```

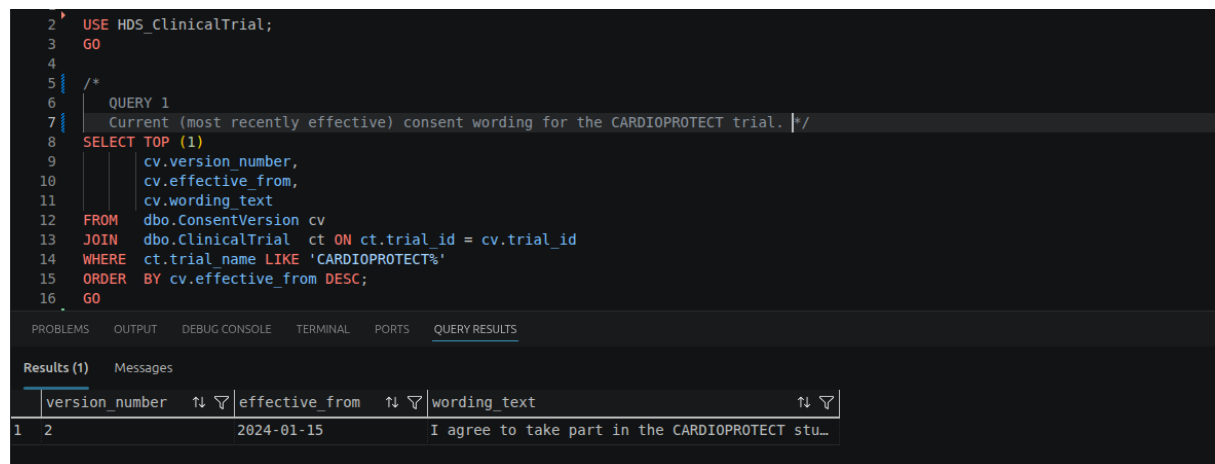
2.2 SQL Queries

2.2.1 Query 1 — Current consent wording for CARDIOPROTECT

Clinical question. Before a consent discussion, a research nurse needs the current (most recently effective) consent wording for CARDIOPROTECT: version number, effective date, and full wording.

```
SELECT TOP (1) cv.version_number, cv.effective_from, cv.wording_text
FROM    dbo.ConsentVersion cv
JOIN    dbo.ClinicalTrial ct ON ct.trial_id = cv.trial_id
WHERE   ct.trial_name LIKE 'CARDIOPROTECT%'
ORDER  BY cv.effective_from DESC;
```

Screenshot.



The screenshot shows the VS Code mssql extension interface. The top pane displays the SQL query for Query 1, which is identical to the one in the previous block. The bottom pane shows the query results in a table grid. The table has three columns: version_number, effective_from, and wording_text. The first row of results shows version 2, effective date 2024-01-15, and the wording 'I agree to take part in the CARDIOPROTECT stu...'.

version_number	effective_from	wording_text
2	2024-01-15	I agree to take part in the CARDIOPROTECT stu...

Figure 2.1: Query 1 output grid (VS Code mssql extension, SQL Server 2025).

Expected result: version 2, effective 2024-01-15, the wearable-devices wording.

Discussion. The query filters to CARDIOPROTECT, orders versions by `effective_from` descending and keeps the top row, so it always returns the latest version even after future amendments. *Data quality:* it relies on `effective_from` being accurate — a new version entered with a wrong or missing effective date could show the nurse out-of-date wording.

2.2.2 Query 2 — Monthly trial status report

Clinical question. A summary of each trial, its status, how long it has run in days, and whether it has an end date or is ongoing; active trials first, then recruiting, then the rest.

```
SELECT ct.trial_name, ct.status,
       DATEDIFF(DAY, ct.start_date, COALESCE(ct.end_date, CAST(GETDATE() AS
↵ DATE))) AS days_running,
       CASE WHEN ct.end_date IS NULL THEN 'Ongoing' ELSE 'Ended' END AS
↵ end_date_status
FROM    dbo.ClinicalTrial ct
```

```
ORDER BY CASE ct.status WHEN 'active' THEN 1 WHEN 'recruiting' THEN 2 ELSE 3
END,
ct.trial_name;
```

Screenshot.

The screenshot shows a SQL query in the editor and its results in a grid. The query is a monthly status report for clinical trials, ordered by status (active, recruiting, rest). The results grid shows three trials: CARDIOPROTECT (active, Ongoing), BREATHE (recruiting, Ongoing), and MOBILISE (completed, Ended).

	trial_name	status	days_running	end_date_status
1	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	active	1220	Ongoing
2	BREATHE: Pulmonary Rehabilitation in COPD	recruiting	762	Ongoing
3	MOBILISE: Early Mobilisation Following Hip Replacement	completed	1460	Ended

Figure 2.2: Query 2 output grid (VS Code mssql extension, SQL Server 2025).

Expected result (3 rows): CARDIOPROTECT (active, Ongoing) first; BREATHE (recruiting, Ongoing) second; MOBILISE (completed, Ended, ~1460 days) last.

Discussion. COALESCE substitutes today’s date for trials with no end date so “days running” is meaningful for ongoing trials, and a CASE expression in ORDER BY gives the custom status priority. *Data quality:* “days running” for ongoing trials changes every day the report is run, so it is only a dated snapshot.

2.2.3 Query 3 — Trial history for participant WBK-003

Clinical question. Full history for WBK-003: every trial, enrolment date, inactivation date, and reason code; ordered by enrolment date. Active enrolments have no reason.

```
SELECT ct.trial_name, e.enrolment_date, e.inactivation_date, ir.reason_code
FROM    dbo.Enrollment      e
JOIN    dbo.Participant      p ON p.participant_id = e.participant_id
JOIN    dbo.TrialSite        ts ON ts.trial_site_id = e.trial_site_id
JOIN    dbo.ClinicalTrial    ct ON ct.trial_id      = ts.trial_id
LEFT JOIN dbo.InactivationReason ir ON ir.reason_id =
e.inactivation_reason_id
```

```
WHERE p.study_code = 'WBK-003'
ORDER BY e.enrolment_date;
```

Screenshot.

The screenshot shows a SQL query in the editor and its results in a table grid. The query is a LEFT JOIN between Enrollment, Participant, TrialSite, ClinicalTrial, and InactivationReason tables, filtered by study_code = 'WBK-003' and ordered by enrolment_date.

Query 3: Full trial history for participant WBK-003. A LEFT JOIN to InactivationReason keeps active enrolments (which have no reason) in the result set. */

```
SELECT ct.trial_name,
       e.enrolment_date,
       e.inactivation_date,
       ir.reason_code
FROM   dbo.Enrollment e
JOIN   dbo.Participant p ON p.participant_id = e.participant_id
JOIN   dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
JOIN   dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
LEFT JOIN dbo.InactivationReason ir ON ir.reason_id = e.inactivation_reason_id
WHERE  p.study_code = 'WBK-003'
ORDER BY e.enrolment_date;
```

Results (1)

	trial_name	enrolment_date	inactivation_date	reason_code
1	BREATHE: Pulmonary Rehabilitation in COPD	2024-06-10	2024-09-15	consent_withdrawn
2	CARDIOPROTECT: Cardiac Rehabilitation in Post...	2024-10-01	NULL	NULL

Figure 2.3: Query 3 output grid (VS Code mssql extension, SQL Server 2025).

Expected result (2 rows): BREATHE, 2024-06-10, 2024-09-15, consent_withdrawn; then CARDIOPROTECT, 2024-10-01, NULL, NULL.

Discussion. A LEFT JOIN to InactivationReason is essential: an active enrolment has a NULL reason and an INNER join would silently drop it. *Data quality:* a participant who left but whose inactivation_reason_id was never populated would show a leaving date with a blank reason — the paired-null check constraint prevents exactly this.

2.2.4 Query 4 — Audit of currently active enrolments

Clinical question. Each currently **active** participant with their trial, site, and enrolment date; ordered by trial name then enrolment date.

```
SELECT p.study_code, ct.trial_name, hs.site_name, e.enrolment_date
FROM   dbo.Enrollment e
JOIN   dbo.Participant p ON p.participant_id = e.participant_id
JOIN   dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
JOIN   dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
JOIN   dbo.HospitalSite hs ON hs.site_id = ts.site_id
WHERE  e.inactivation_date IS NULL
ORDER BY ct.trial_name, e.enrolment_date;
```

Screenshot.

```

78
79
80 USE HDS_ClinicalTrial;
81 GO
82 /*
83  QUERY 4: Audit of currently ACTIVE enrolments: participant, trial, site, enrol date.
84  Order by trial name then enrolment date. */
85 SELECT p.study_code,
86        ct.trial_name,
87        hs.site_name,
88        e.enrolment_date
89 FROM    dbo.Enrollment e
90 JOIN    dbo.Participant p ON p.participant_id = e.participant_id
91 JOIN    dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
92 JOIN    dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
93 JOIN    dbo.HospitalSite hs ON hs.site_id = ts.site_id
94 WHERE   e.inactivation_date IS NULL
95 ORDER  BY ct.trial_name, e.enrolment_date;
96 GO
97
98
99
100
101
102

```

Results (1) Messages

	study_code	trial_name	site_name	enrolment_date
1	WBK-005	BREATHE: Pulmonary Rehabilitation in COPD	Westbrook General Hospital	2024-08-05
2	WBK-001	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	Westbrook General Hospital	2023-03-15
3	WBK-003	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	Westbrook General Hospital	2024-10-01
4	WBK-002	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	St Katherine's Hospital	2025-01-10

Figure 2.4: Query 4 output grid (VS Code mssql extension, SQL Server 2025).

Expected result (4 rows): WBK-005 / BREATHE / Westbrook General / 2024-08-05; WBK-001 / CARDIOPROTECT / Westbrook General / 2023-03-15; WBK-003 / CARDIOPROTECT / Westbrook General / 2024-10-01; WBK-002 / CARDIOPROTECT / St Katherine's / 2025-01-10.

Discussion. “Active” is expressed as `inactivation_date IS NULL`; joining through `TrialSite` to both the trial and the site reconstructs where each active participant sits. *Data quality:* the definition of “active” depends on inactivation being recorded promptly — a participant who has left but whose exit was not entered would wrongly appear here.

2.2.5 Query 5 — Enrolment count per site (including empty sites)

Clinical question. How many participants have ever been enrolled at each site, including sites with none. Show site name, city, total count, descending.

```

SELECT hs.site_name, hs.city, COUNT(e.enrollment_id) AS total_enrolments
FROM    dbo.HospitalSite hs
LEFT JOIN dbo.TrialSite ts ON ts.site_id = hs.site_id
LEFT JOIN dbo.Enrollment e ON e.trial_site_id = ts.trial_site_id
GROUP BY hs.site_name, hs.city
ORDER BY total_enrolments DESC;

```

Screenshot.

```

99
100
101 USE HDS_ClinicalTrial;
102 GO
103
104 /*
105  QUERY 5: Total enrolments ever, per site, INCLUDING sites with none.
106  LEFT JOIN from HospitalSite preserves sites with zero enrolments;
107  COUNT(e.enrollment_id) counts rows, not NULLs, so empty sites score 0.*/
108 SELECT hs.site_name,
109        hs.city,
110        COUNT(e.enrollment_id) AS total_enrolments
111 FROM    dbo.HospitalSite hs
112 LEFT JOIN dbo.TrialSite ts ON ts.site_id = hs.site_id
113 LEFT JOIN dbo.Enrollment e ON e.trial_site_id = ts.trial_site_id
114 GROUP BY hs.site_name, hs.city
115 ORDER BY total_enrolments DESC;
116 GO
117
118
119
120
121
122
123

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	site_name	city	total_enrolments
1	Westbrook General Hospital	London	4
2	St Katherine's Hospital	Manchester	3
3	Ridgeway Community Hospital	Bristol	0

Figure 2.5: Query 5 output grid (VS Code mssql extension, SQL Server 2025).

Expected result (3 rows): Westbrook General (London) 4; St Katherine's (Manchester) 3; Ridgeway Community (Bristol) 0.

Discussion. Starting from `HospitalSite` with `LEFT JOINs` preserves sites with no enrolments, and `COUNT(e.enrollment_id)` counts rows rather than `NULLs` so empty sites correctly score 0. *Data quality:* a site that participates in a trial but has not yet recorded any enrolments is indistinguishable from one that genuinely has none — the zero needs interpretation in context.

2.2.6 Query 6 — Trials with an amended protocol

Clinical question. Trials where the protocol was amended (more than one consent version): trial name, current status, number of versions; descending; only where more than one version exists.

```

SELECT ct.trial_name, ct.status, COUNT(cv.consent_version_id) AS version_count
FROM    dbo.ClinicalTrial ct
JOIN    dbo.ConsentVersion cv ON cv.trial_id = ct.trial_id
GROUP BY ct.trial_name, ct.status
HAVING COUNT(cv.consent_version_id) > 1
ORDER BY version_count DESC;

```

Screenshot.

```

121
122
123
124 USE HDS_ClinicalTrial;
125 GO
126 /*
127  QUERY 6: Trials with an amended protocol (more than one consent version issued). */
128 SELECT ct.trial_name,
129        ct.status,
130        COUNT(cv.consent_version_id) AS version_count
131 FROM    dbo.ClinicalTrial ct
132 JOIN    dbo.ConsentVersion cv ON cv.trial_id = ct.trial_id
133 GROUP BY ct.trial_name, ct.status
134 HAVING COUNT(cv.consent_version_id) > 1
135 ORDER BY version_count DESC;
136 GO
137
138
139
140
141
142
143
144
145

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Results (1) Messages

	trial_name	status	version_count
1	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	active	2

Figure 2.6: Query 6 output grid (VS Code mssql extension, SQL Server 2025).

Expected result (1 row): CARDIOPROTECT, active, 2.

Discussion. Grouping by trial and counting versions, then filtering with `HAVING COUNT(...) > 1`, isolates amended trials (`HAVING` filters the aggregate, which `WHERE` cannot). *Data quality:* this counts version *rows*, so a duplicate version entered in error would overstate the number of genuine amendments.

2.2.7 Query 7 — Most recently enrolled participant per site

Clinical question. The most recently enrolled participant at each site across all trials: study code, trial name, site name, enrolment date.

```

WITH RankedEnrolments AS (
    SELECT p.study_code, ct.trial_name, hs.site_name, e.enrolment_date,
           ROW_NUMBER() OVER (PARTITION BY hs.site_id
                               ORDER BY e.enrolment_date DESC, e.enrollment_id
                               DESC) AS rn
    FROM    dbo.Enrollment e
    JOIN    dbo.Participant p ON p.participant_id = e.participant_id
    JOIN    dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
    JOIN    dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
    JOIN    dbo.HospitalSite hs ON hs.site_id = ts.site_id
)
SELECT study_code, trial_name, site_name, enrolment_date
FROM    RankedEnrolments
WHERE   rn = 1

```

```
ORDER BY site_name;
```

Screenshot.

The screenshot shows a SQL query in the VS Code mssql extension. The query is designed to find the most recently enrolled participant at each site across all trials. It uses a CTE named 'RankedEnrolments' to rank participants by enrolment date descending, with 'enrollment_id' as a tie-breaker. The final result is ordered by site name.

Query 7:

```

151 USE HDS_ClinicalTrial;
152 GO
153 /* QUERY 7: Most recently enrolled participant at each site (across all trials).
154    ROW_NUMBER partitioned by site, newest enrolment first; keep rank 1. */
155 WITH RankedEnrolments AS (
156     SELECT p.study_code,
157            ct.trial_name,
158            hs.site_name,
159            e.enrolment_date,
160            ROW_NUMBER() OVER (PARTITION BY hs.site_id
161                               ORDER BY e.enrolment_date DESC, e.enrollment_id DESC) AS rn
162     FROM   dbo.Enrollment e
163     JOIN   dbo.Participant p ON p.participant_id = e.participant_id
164     JOIN   dbo.TrialSite ts ON ts.trial_site_id = e.trial_site_id
165     JOIN   dbo.ClinicalTrial ct ON ct.trial_id = ts.trial_id
166     JOIN   dbo.HospitalSite hs ON hs.site_id = ts.site_id
167 )
168 SELECT study_code,
169        trial_name,
170        site_name,
171        enrolment_date
172 FROM   RankedEnrolments
173 WHERE  rn = 1
174 ORDER BY site_name;
175 GO
176

```

Query Results:

	study_code	trial_name	site_name	enrolment_date
1	WBK-002	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	St Katherine's Hospital	2025-01-10
2	WBK-003	CARDIOPROTECT: Cardiac Rehabilitation in Post-MI Patients	Westbrook General Hospital	2024-10-01

Figure 2.7: Query 7 output grid (VS Code mssql extension, SQL Server 2025).

Expected result (2 rows): WBK-002 / CARDIOPROTECT / St Katherine’s Hospital / 2025-01-10; WBK-003 / CARDIOPROTECT / Westbrook General Hospital / 2024-10-01. (Ridgeway has no enrolments, so it does not appear.)

Discussion. ROW_NUMBER() partitioned by site and ordered by enrolment date descending ranks each site’s enrolments; keeping `rn = 1` returns the newest per site, with `enrollment_id` as a deterministic tie-breaker. *Data quality:* if two participants share the same enrolment date the tie-break is arbitrary, so “most recent” may need a finer timestamp than a date to be unambiguous.

Chapter 3

Part 3: Reflection

3.1 Why no participant names?

Storing only a `study_code` pseudonymises the registry: it applies the UK GDPR data-minimisation principle (Article 5(1)(c)) and the Caldicott Principles of justifying every use of confidential information and using the minimum necessary, and means a breach of the trial database does not directly expose identities. Direct identifiers such as names live in a separate, access-controlled master linkage list held by the research office, not in the analytical database. In practice this shapes how staff work. Researchers query and report entirely on `study_code`, so trial analyses can be shared more freely because they carry no obvious identifiers. Clinical staff who need to act on a real patient — to contact them or update their medical record — must re-identify the person through the controlled linkage list, which is deliberately a separate, auditable step. The cost is friction and a dependency on that linkage being accurate and available: a broken or out-of-date mapping could mean a participant cannot be reached for a safety issue. The design therefore trades day-to-day convenience for a much stronger confidentiality and information-governance position.

3.2 A situation where the data could mislead

Consider Query 4 and Query 5, which both depend on `inactivation_date`. If a participant has actually left a trial but their exit has not yet been entered, they will still appear as “active” and will be counted in active-enrolment reports — overstating recruitment and potentially prompting unnecessary follow-up contact. The data is not *wrong* field by field; it is *stale*, which is harder to spot. A data manager could detect this by reconciling the registry against source records: cross-checking active enrolments against clinic attendance or against the standardised `InactivationReason` log, and flagging participants with no recent activity (e.g. no consent or visit events for an unusually long period) for review. Preventively, they could add a periodic data-quality report listing long-inactive “active” enrolments, set a service standard for how quickly exits must be recorded, and use the `lost_to_follow_up` reason code consistently so genuine non-contact is captured rather than left as a silent gap. The fix is process plus monitoring, not just a query.

3.3 Why consent records are append-only, and how to enforce it

Consent records are the legal and ethical evidence that each participant agreed to the specific protocol version in force at the time. They must be append-only because consent is a historical fact: Good Clinical Practice (ICH-GCP) requires trial-critical records to carry a complete, tamper-evident audit trail, and editing or deleting a signing event would destroy the evidence that regulators, sponsors and ethics committees rely on to prove valid consent existed. If the rule were broken, a re-consent after a protocol amendment could be silently overwritten or a withdrawn consent erased, making it impossible to demonstrate which wording a participant actually agreed to and undermining every downstream analysis that assumes valid consent.

I would enforce this at the database level with an **INSTEAD OF UPDATE, DELETE trigger** on `ConsentSigning` that rejects any such operation with an error, while leaving `INSERT` free (see `05_append_only_trigger.sql`):

```
CREATE OR ALTER TRIGGER dbo.trg_ConsentSigning_AppendOnly
ON dbo.ConsentSigning
INSTEAD OF UPDATE, DELETE
AS
BEGIN
    SET NOCOUNT ON;
    THROW 51000, 'ConsentSigning is append-only: rows cannot be updated or
    ↪ deleted.', 1;
END;
```

This guarantees the rule regardless of which application or user connects, which a permission grant alone would not. For stronger assurance, a SQL Server append-only **ledger table** refuses `UPDATE/DELETE` at the engine level and chains rows into a cryptographically verifiable hash tree, so even an administrator cannot tamper with a consent record undetectably — stronger than the trigger, which a sufficiently privileged account could disable.

Appendix A — Data dictionary

A complete column-level reference for every table, derived from `script/01_create_tables.sql`.
Key: PK = primary key, FK = foreign key, UK = unique (business) key. All surrogate *_id keys are INT IDENTITY(1,1).

This dictionary is not static documentation: `script/06_metadata.sql` stores every table description — and the column descriptions where meaning is not self-evident, in particular the meaningful-NULL columns — **in the database catalog itself** as MS_Description extended properties. The dictionary can therefore be regenerated at any time (or consumed by data-catalogue tooling) by querying the system views, so the metadata travels with the schema rather than living in a document that can drift out of date:

```
SELECT t.name AS table_name,
       c.name AS column_name,
       TYPE_NAME(c.user_type_id) AS data_type,
       IIF(c.is_nullable = 1, 'NULL', 'NOT NULL') AS nullability,
       CAST(ep.value AS NVARCHAR(400)) AS column_description
FROM sys.tables t
JOIN sys.columns c ON c.object_id = t.object_id
LEFT JOIN sys.extended_properties ep
      ON ep.major_id = c.object_id AND ep.minor_id = c.column_id
      AND ep.class = 1 AND ep.name = 'MS_Description'
ORDER BY t.name, c.column_id;
```

ClinicalTrial

Table 3.1: ClinicalTrial

Column	Type	Null	Key	Notes
trial_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
registration_number	VARCHAR(20)	NOT NULL	UK	ISRCTN number; externally recognised natural key.
trial_name	NVAR- CHAR(150)	NOT NULL		Full trial title.
phase	VARCHAR(4)	NOT NULL		CHECK IN ('I','II','III','IV').
status	VARCHAR(20)	NOT NULL		CHECK IN ('recruiting','active','completed','suspended').

Column	Type	Null	Key	Notes
<code>start_date</code>	DATE	NOT NULL		Trial start date.
<code>end_date</code>	DATE	NULL		NULL = ongoing. CHECK (<code>end_date</code> >= <code>start_date</code>).

HospitalSite

Table 3.2: HospitalSite

Column	Type	Null	Key	Notes
<code>site_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>site_code</code>	VARCHAR(10)	NOT NULL	UK	Short operational code, e.g. WBK-GEN .
<code>site_name</code>	NVAR- CHAR(100)	NOT NULL		Hospital site name.
<code>city</code>	NVAR- CHAR(60)	NOT NULL		City.
<code>country</code>	NVAR- CHAR(60)	NOT NULL		Country.

Participant

Table 3.3: Participant

Column	Type	Null	Key	Notes
<code>participant_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>study_code</code>	VAR- CHAR(15)	NOT NULL	UK	Pseudonym; only external identifier (no names held).
<code>date_of_birth</code>	DATE	NOT NULL		CHECK (<code>date_of_birth</code> < <code>registration_date</code>).
<code>sex_at_birth</code>	VAR- CHAR(10)	NOT NULL		CHECK IN ('Male', 'Female', 'Intersex', 'Unknown').
<code>registration_date</code>	DATE	NOT NULL		Date first registered with the research office.

InactivationReason

Table 3.4: InactivationReason

Column	Type	Null	Key	Notes
<code>reason_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>reason_code</code>	VARCHAR(30)	NOT NULL	UK	Standardised report token, e.g. consent_withdrawn .

Column	Type	Null	Key	Notes
reason_description	NVAR-CHAR(255)	NOT NULL		Human-readable description of the reason.

TrialSite (link table)

Table 3.5: TrialSite

Column	Type	Null	Key	Notes
trial_site_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
trial_id	INT	NOT NULL	FK, UK	-> ClinicalTrial. Part of UNIQUE (trial_id, site_id).
site_id	INT	NOT NULL	FK, UK	-> HospitalSite. Part of UNIQUE (trial_id, site_id).
opened_date	DATE	NOT NULL		Date the site opened for recruitment on this trial.
closed_date	DATE	NULL		NULL = still open. CHECK (closed_date >= opened_date).

ConsentVersion

Table 3.6: ConsentVersion

Column	Type	Null	Key	Notes
consent_version_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
trial_id	INT	NOT NULL	FK, UK	-> ClinicalTrial. Part of UNIQUE (trial_id, version_number).
version_number	INT	NOT NULL	UK	CHECK (version_number > 0). Part of the unique key.
effective_from	DATE	NOT NULL		Date this version came into effect.
wording_text	NVAR-CHAR(MAX)	NOT NULL		Full consent-form wording; never overwritten.

Enrollment

Table 3.7: Enrollment. A filtered unique index allows only one active enrolment per participant (WHERE inactivation_date IS NULL).

Column	Type	Null	Key	Notes
enrollment_id	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
participant_id	INT	NOT NULL	FK, UK	-> Participant. Part of UNIQUE (participant_id, trial_site_id).
trial_site_id	INT	NOT NULL	FK, UK	-> TrialSite (captures trial and site). Part of the unique key.

Column	Type	Null	Key	Notes
<code>enrolment_date</code>	DATE	NOT NULL		Date enrolment began.
<code>inactivation_date</code>	DATE	NULL		NULL = still active. CHECK (<code>inactivation_date >= enrolment_date</code>).
<code>inactivation_reason_id</code>	INT	NULL	FK	-> <code>InactivationReason</code> . Paired-null CHECK with <code>inactivation_date</code> .

ConsentSigning (append-only)

Table 3.8: ConsentSigning. Protected by an **INSTEAD OF UPDATE, DELETE** trigger so rows can only ever be inserted.

Column	Type	Null	Key	Notes
<code>consent_signing_id</code>	INT IDENTITY	NOT NULL	PK	Surrogate primary key.
<code>enrollment_id</code>	INT	NOT NULL	FK, UK	-> <code>Enrollment</code> . Part of UNIQUE (<code>enrollment_id, consent_version_id</code>).
<code>consent_version_id</code>	INT	NOT NULL	FK, UK	-> <code>ConsentVersion</code> . Part of the unique key.
<code>signed_date</code>	DATE	NOT NULL		Date the consent was signed.
<code>witnessed_by</code>	NVAR- CHAR(100)	NOT NULL		Name/role of the witnessing staff member.

Appendix B — File manifest

Table 3.9: File manifest

File	Contents
script/00_create_database.sql	(Re)creates the HDS_ClinicalTrial database
script/01_create_tables.sql	CREATE TABLEs, constraints, indexes
script/02_insert_reference_data.sql	The five reference sheets from the brief
script/03_insert_operational_data.sql	TrialSite, Enrollment, ConsentSigning data
script/04_queries.sql	The seven queries
script/05_append_only_trigger.sql	Append-only enforcement for ConsentSigning
script/06_metadata.sql	Extended-property metadata + catalog-driven data dictionary
script/HDS_DB_LAINSBURY_2026.sql	Submission file — everything in run order

Appendix C — Interoperability mapping (HL7 FHIR)

The registry does not exist in isolation: if the trust later needs to exchange trial data with other NHS organisations, sponsors, or national research platforms, the natural target is the HL7 FHIR standard used across the NHS. The schema maps cleanly onto FHIR resources, which validates the entity choices and gives a ready-made route for future data linkage beyond the trust:

Table 3.10: Mapping of registry tables to HL7 FHIR resources

Registry table	FHIR resource	Notes
ClinicalTrial	ResearchStudy	Phase and status map to the resource's phase and status elements.
Participant	Patient (pseudonymised)	study_code becomes a pseudonymous identifier ; no name elements populated.
Enrollment	ResearchSubject	FHIR likewise links one patient to one study with period and status.
HospitalSite	Location / Organization	Site code as the identifier .
ConsentSigning + ConsentVersion	Consent	FHIR Consent also references the specific policy/form version agreed to.
InactivationReason	ResearchSubject progress reason (coded)	The controlled vocabulary would be published as a FHIR CodeSystem .

Coversheet

MSt Healthcare Data Science — Authentication of practice

I confirm that I have fully read and understood the assignment brief for this module. **Y**

Details

Name	Malik J. Lainsbury
Submission Date	03 – 07 – 26
Word Count: whole assignment including codes	~2,900 word-equivalents (1,447 prose + 7 queries x 150 + 8 INSERT statements x 50)
Word Count: main body excluding abstract, references and supplementary materials	1,447 prose words (appendices excluded as supplementary)

Permission to share your assignment

I do give permission to share my assignment with future MSt participants.

University statement of originality

This assignment is the result of my own work and includes nothing which is the outcome of work done in collaboration except as declared in the Preface and specified in the text. It is not substantially the same as any that I have previously submitted for a degree or diploma or other qualification at the University of Cambridge or any other university or similar institution, or that is being concurrently submitted, except as declared in the Preface and specific in the text. I further state that no substantial part of my Portfolio has already been submitted, nor is being concurrently submitted for any such degree, diploma or other qualification at the University of

Cambridge or any other university or similar institution except as declared in the Preface and specified in the text.

I confirm the statement of originality as above **Y**

Questions for reflection

1. Which aspects of this assignment are you most uncertain about and/or would most like to receive feedback on? Whether pointing `Enrollment` at the `TrialSite` link table (rather than holding separate trial and site foreign keys) is considered good practice or over-normalisation; and whether the filtered unique index is an acceptable way to express the “one active trial at a time” rule.

5. Using the wording in the rubric, how would you describe the quality of the different aspects of your work? The ERD, table descriptions and relationships aim at the “high performance” descriptors: all entities identified, the many-to-many resolved with a link table, and each decision justified against the scenario. The script runs end-to-end without errors and every foreign key is resolved by subquery on a business key.

Use of generative artificial intelligence

Which permitted use of generative AI are you acknowledging?	Review and verification of my own draft work
Which generative AI tool did you use (name and version)?	Claude Code (Anthropic Claude Fable 5)
What did you use the tool for?	Reviewing the draft report and SQL script against the brief; executing the SQL script on Microsoft SQL Server to confirm it runs without errors and that the seven queries return the expected results; checking internal consistency between the report and the script; computing the coversheet word counts.
How have you used or changed the generative AI’s output?	The database design, SQL and written prose are my own. Following the AI-assisted review I corrected one inconsistency (aligning the <code>sex_at_birth</code> CHECK constraint in the combined script with the data dictionary) and completed the coversheet.
